

SQL Date & Time Notes

Internal Storage

Users see readable dates.

Internally, databases store dates in optimized binary formats for efficient comparison and arithmetic.

Unix Timestamp

Counts the number of seconds since the Unix epoch.

Epoch	1970-01-01 00:00:00 UTC
Example	1752887730 represents a specific moment in time.

Epoch Time Advantages

- Easy arithmetic
- Compact storage
- Fast comparisons

Fractional Seconds

Many databases support milliseconds or microseconds.

Example: **2026-07-19 14:35:28.123456**

NOW()

Returns the current date and time.

MySQL

```
SELECT NOW();
```

Example result: **2026-07-19 14:35:28**

| CURRENT_TIME

Returns only the current time.

```
SELECT CURRENT_TIME;
```

| CURRENT_TIMESTAMP

ANSI SQL standard function.

```
SELECT CURRENT_TIMESTAMP;
```

Result: **2026-07-19 14:35:28**

| LOCALTIMESTAMP

Returns the current local timestamp.
Common in PostgreSQL.

```
SELECT LOCALTIMESTAMP;
```

In MySQL NOW() and CURRENT_TIMESTAMP return the same value.

| Adding Days

MySQL

```
SELECT DATE_ADD('2026-07-19', INTERVAL 10 DAY);
```

↓ **2026-07-29**

3.3 Subtracting Days

```
SELECT DATE_SUB('2026-07-19', INTERVAL 7 DAY);
```

↓ **2026-07-12**

3.4 Adding Weeks

```
SELECT DATE_ADD('2026-07-19', INTERVAL 2 WEEK);
```

↓ 2026-08-02

3.5 Adding Months

```
SELECT DATE_ADD('2026-07-19', INTERVAL 3 MONTH);
```

↓ 2026-10-19

3.6 Adding Years

```
SELECT DATE_ADD('2026-07-19', INTERVAL 5 YEAR);
```

↓ 2031-07-19

| DATEADD()

Important: DATEADD() is not available in MySQL. It is used in SQL Server. MySQL uses DATE_ADD()

DATEADD() adds a specified interval to a date or time.

4.2 Syntax (SQL Server)

```
DATEADD(datepart, number, date)
```

Example

```
SELECT DATEADD(DAY, 10, '2026-07-19');
```

↓ 2026-07-29

Vendor Alternatives

Database	Equivalent
MySQL	DATE_ADD(date, INTERVAL n unit)

Syntax

MySQL

```
DATE_ADD(date, INTERVAL value unit)
```

or

```
date + INTERVAL value unit
```

```
SELECT NOW() + INTERVAL 3 HOUR;  
SELECT NOW() + INTERVAL 30 MINUTE;
```

Example: OTP valid for 30 minutes.

Mixed Intervals

Different vendors support combining intervals differently.

MySQL

```
SELECT DATE_ADD(  
    DATE_ADD(CURRENT_DATE, INTERVAL 1 YEAR),  
    INTERVAL 2 MONTH  
);
```

PostgreSQL

```
SELECT CURRENT_DATE + INTERVAL '1 year 2 months';
```

Introduction to AGE()

AGE() is a PostgreSQL function.

It returns the elapsed time between two dates.

Unlike DATEDIFF(), it preserves years, months and days.

6.2 Syntax

```
AGE(date1, date2)
```

6.3 Age from Birthdate

```
SELECT AGE('2026-07-19', '2003-05-12');
```

↓

23 years
2 months
7 days

AGE vs DATEDIFF

AGE	DATEDIFF
23 years 2 months 5 days	8467 days

Vendor Support

Database	AGE()
PostgreSQL	✓
MySQL	✗
SQL Server	✗

DATEDIFF()

DATEDIFF() calculates the difference between two dates.
Most commonly measured in days.

7.2 Syntax

MySQL

```
DATEDIFF(date1, date2)
```

Result: **date1 - date2**

7.3 Difference in Days

```
SELECT DATEDIFF('2026-07-20','2026-07-15');
```

↓ 5

7.4 Difference in Months

MySQL DATEDIFF does not return months directly.
Instead:

```
TIMESTAMPDIFF(MONTH,start,end)
```

7.5 Difference in Years

Likewise:

```
TIMESTAMPDIFF(YEAR,start,end)
```

7.6 Negative Results

```
SELECT DATEDIFF('2026-07-10','2026-07-20');
```

↓ -10

| TIMESTAMPDIFF()

TIMESTAMPDIFF() returns the difference between two timestamps in a chosen unit.
Unlike DATEDIFF(), you choose the unit.

8.2 Syntax

```
TIMESTAMPDIFF(unit,start,end)
```

8.3 Seconds

```
SELECT  
TIMESTAMPDIFF(SECOND,  
login_time,  
logout_time  
);
```